

Unit 112 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 80.5 ac. 40 plots, BAF 20, variable-radius cruise on a 300 ft grid, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 2.5 %)	Type	Sierran mixed conifer
Trees per acre	92	CWHR size / density	4 / D (cover 87 %)
QMD	20.6 in	SDI	279 (48 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	210 sq ft/ac 21.5 in	BA QMD	4 sq ft/ac 8.6 in
TPA	84	TPA	9
Volume	33.2 MBF/ac (6608 cu ft)	Volume	1.4 green tons/ac (64 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 156 sq ft/ac BA	Vol. sampling error	9.3 % at 95 % (CV 29 %, t = 2.02, n = 40)
Species (BA)	WF 66, PP 63, IC 36, DF 30, SP 19	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	4.1 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	5.1 % at 95 % (CV 16 %); reference, not the standard
		Measured trees	PP 126, WF 131, DF 61, SP 38, IC 72; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	40; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	2	3.4	1.0	15	0.1
8-11.9	19	16.0	9.5	206	1.0
12-15.9	47	21.9	23.5	523	2.6
16-19.9	62	17.6	31.0	787	3.9
20-23.9	61	11.7	30.5	865	4.3
24-27.9	69	9.5	34.5	1112	5.6
28-31.9	51	5.2	25.5	869	4.4
32-35.9	42	3.4	21.0	735	3.7
36-39.9	28	1.8	14.0	566	2.8
40-43.9	12	0.6	6.0	235	1.2
44-47.9	12	0.5	6.0	246	1.2
48-51.9	9	0.3	4.5	199	1.0
52-55.9	9	0.3	4.5	202	1.0
56-59.9	1	0.0	0.5	23	0.1
60-63.9	4	0.1	2.0	88	0.4
All	428	92.5	214.0	6672	33.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	131	27.0	65.5	21.1	2068	10.4
PP	126	27.2	63.0	20.6	2148	10.8
IC	72	15.3	36.0	20.7	966	4.8
DF	61	14.7	30.5	19.5	904	4.5
SP	38	8.3	19.0	20.5	585	2.9
All	428	92.5	214.0	20.6	6672	33.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112 plots, 300 ft grid (orange: QA flag)

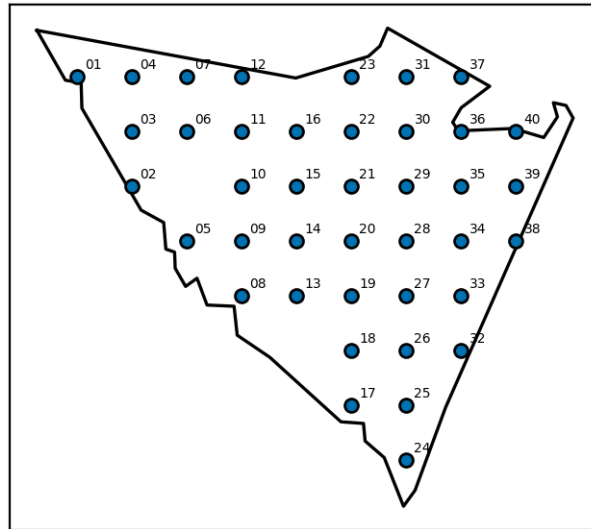


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
112-01	11	220	12	ok
112-02	11	220	10	ok
112-03	11	220	17	ok
112-04	12	240	12	ok
112-05	9	180	21	ok
112-06	12	240	36	ok
112-07	10	200	10	ok
112-08	5	100	21	ok
112-09	12	240	10	ok
112-10	11	220	10	ok
112-11	9	180	10	ok
112-12	11	220	29	ok
112-13	10	200	12	ok
112-14	13	260	8	ok
112-15	13	260	15	ok
112-16	13	260	16	ok
112-17	10	200	13	ok
112-18	12	240	36	ok
112-19	9	180	13	ok
112-20	11	220	21	ok
112-21	10	200	14	ok
112-22	13	260	13	ok
112-23	10	200	22	ok
112-24	13	260	28	ok
112-25	10	200	26	ok
112-26	12	240	11	ok
112-27	8	160	9	ok
112-28	10	200	15	ok
112-29	8	160	13	ok
112-30	11	220	12	ok
112-31	8	160	7	ok
112-32	12	240	25	ok
112-33	11	220	26	ok
112-34	11	220	20	ok
112-35	10	200	10	ok
112-36	13	260	39	ok
112-37	11	220	16	ok
112-38	9	180	16	ok
112-39	12	240	19	ok
112-40	11	220	30	ok